

ECO101

Introduction to Microeconomics

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Assumptions & Models

- Assumptions simplify the complex world, make it easier to understand.
- Example: To study international trade, assume two countries and two goods.
Unrealistic, but simple to learn and gives useful insights about the real world.
- Model: a highly simplified representation of a more complicated reality.
Economists use models to study economic issues.

Our First Model: The Circular-Flow Diagram

- **The Circular-Flow Diagram:** a visual model of the economy, shows how dollars flow through markets among households and firms
- **Two types of “actors”:**
 - Households
 - Firms
- **Two markets:**
 - The market for goods and services
 - The market for “factors of production”

Factors of Production

The resources the economy uses to produce goods & services, including

- Labor
- Land
- Capital (buildings & machines used in production)

FIGURE 1: The Circular-Flow Diagram

Firms

Firms:

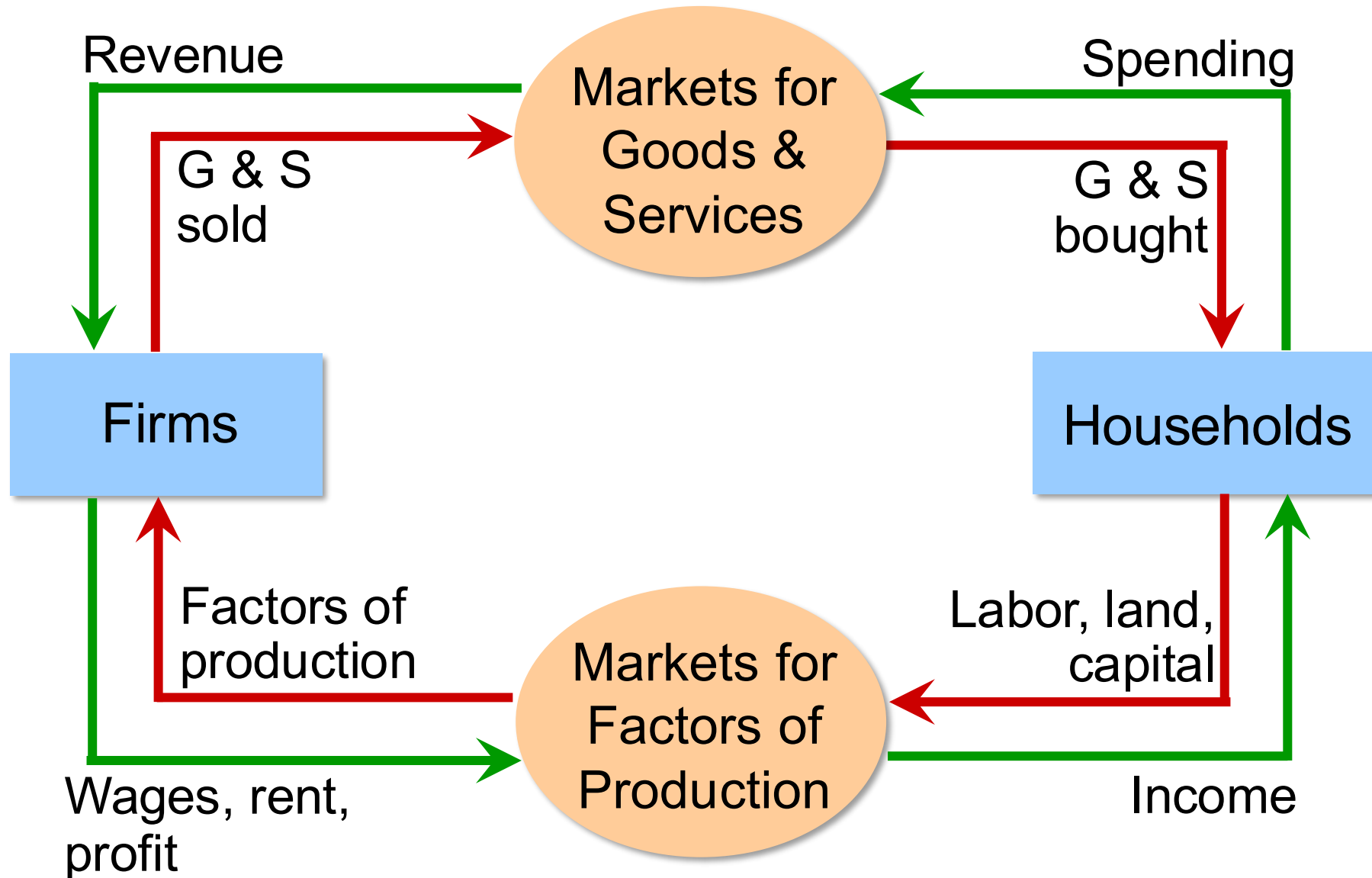
- Buy/hire factors of production, use them to produce goods and services
- Sell goods & services

Households:

- Own the factors of production, sell/rent them to firms for income
- Buy and consume goods & services

Households

FIGURE 1: The Circular-Flow Diagram



Our Second Model: The Production Possibilities Frontier

The **Production Possibilities Frontier (PPF)**:

a graph that shows the combinations of two goods the economy can possibly produce given the available resources and the available technology

Example:

Two goods: computers and wheat

One resource: labor (measured in hours)

Economy has 50,000 labor hours per month available for production.

PPF Example

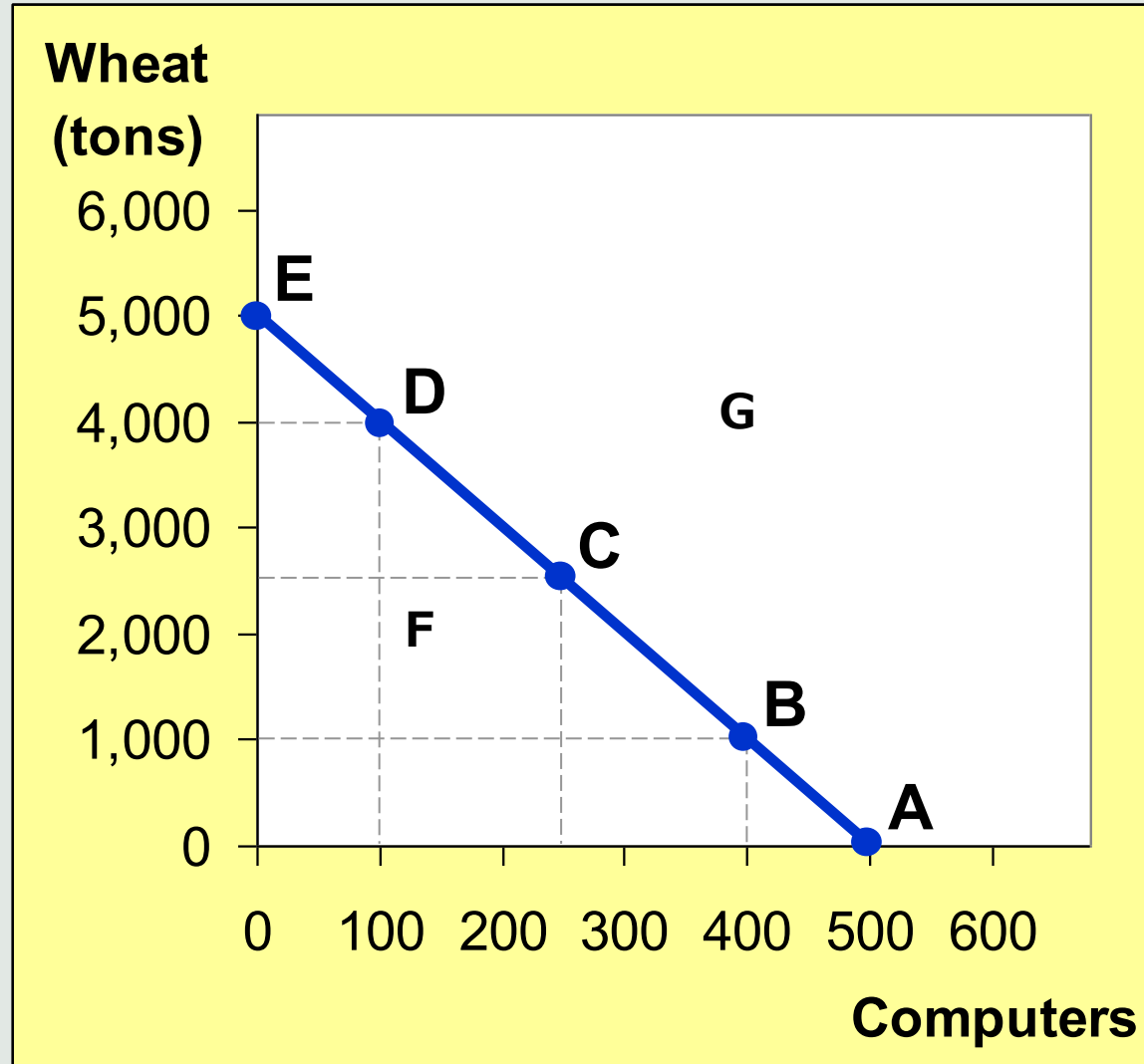
Producing one computer requires 100 hours labor.

Producing one ton of wheat requires 10 hours labor.

	Employment of labor hours		Production	
	Computers	Wheat	Computers	Wheat
A	50,000	0	500	0
B	40,000	10,000	400	1,000
C	25,000	25,000	250	2,500
D	10,000	40,000	100	4,000
E	0	50,000	0	5,000

PPF Example

Point on graph	Production	
	Com- puters	Wheat
A	500	0
B	400	1,000
C	250	2,500
D	100	4,000
E	0	5,000



The PPF: What We Know So Far

Points on the PPF (like A – E)

possible

efficient: all resources are fully utilized

Points under the PPF (like F)

possible

not efficient: some resources underutilized
(e.g., workers unemployed, factories idle)

Points above the PPF (like G)

not possible

The PPF and Opportunity Cost

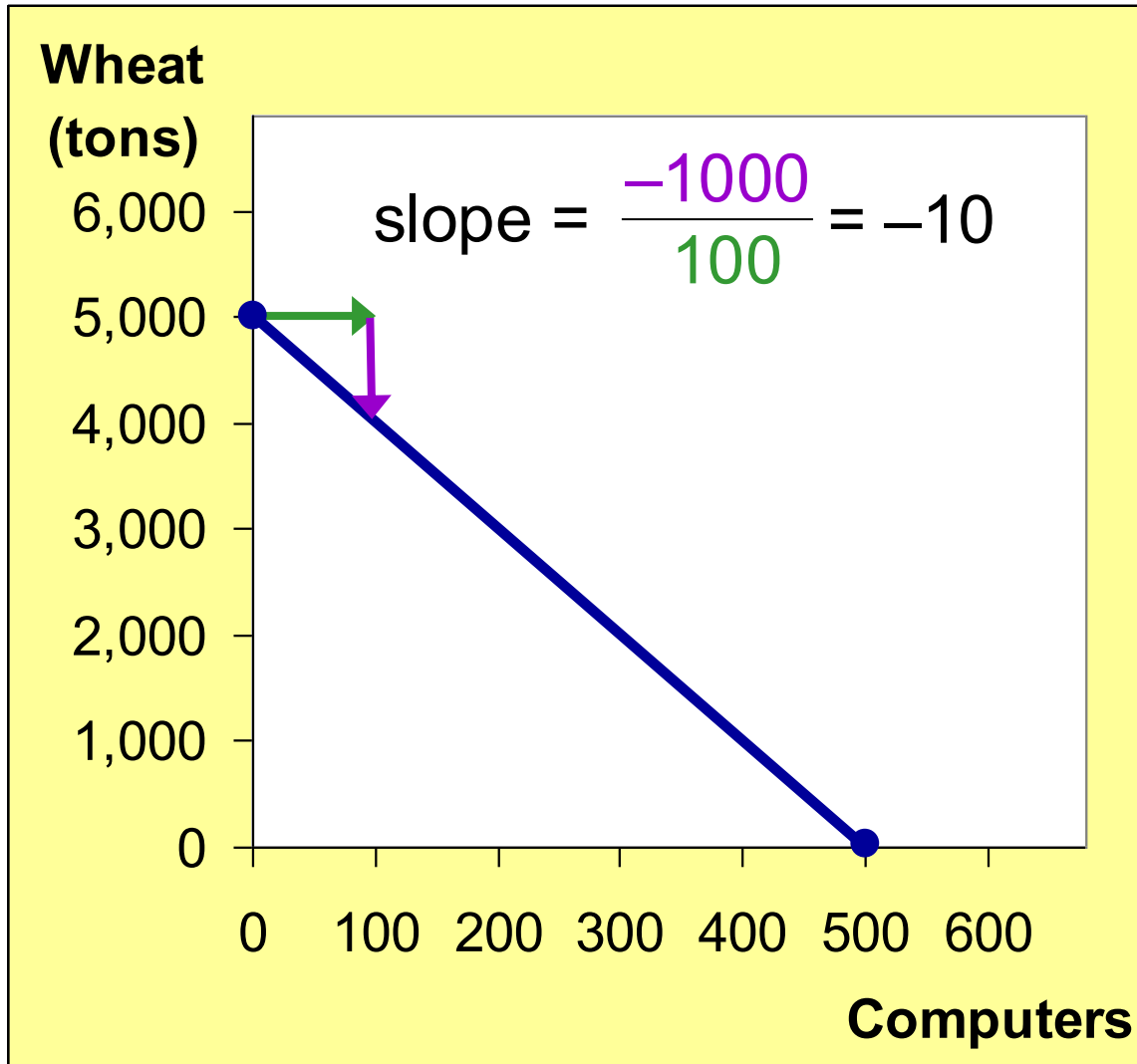
Recall: The **opportunity cost** of an item is what must be given up to obtain that item.

Moving along a PPF involves shifting resources (e.g., labor) from the production of one good to the other.

Society faces a tradeoff: Getting more of one good requires sacrificing some of the other.

The slope of the PPF tells you the opportunity cost of one good in terms of the other.

The PPF and Opportunity Cost

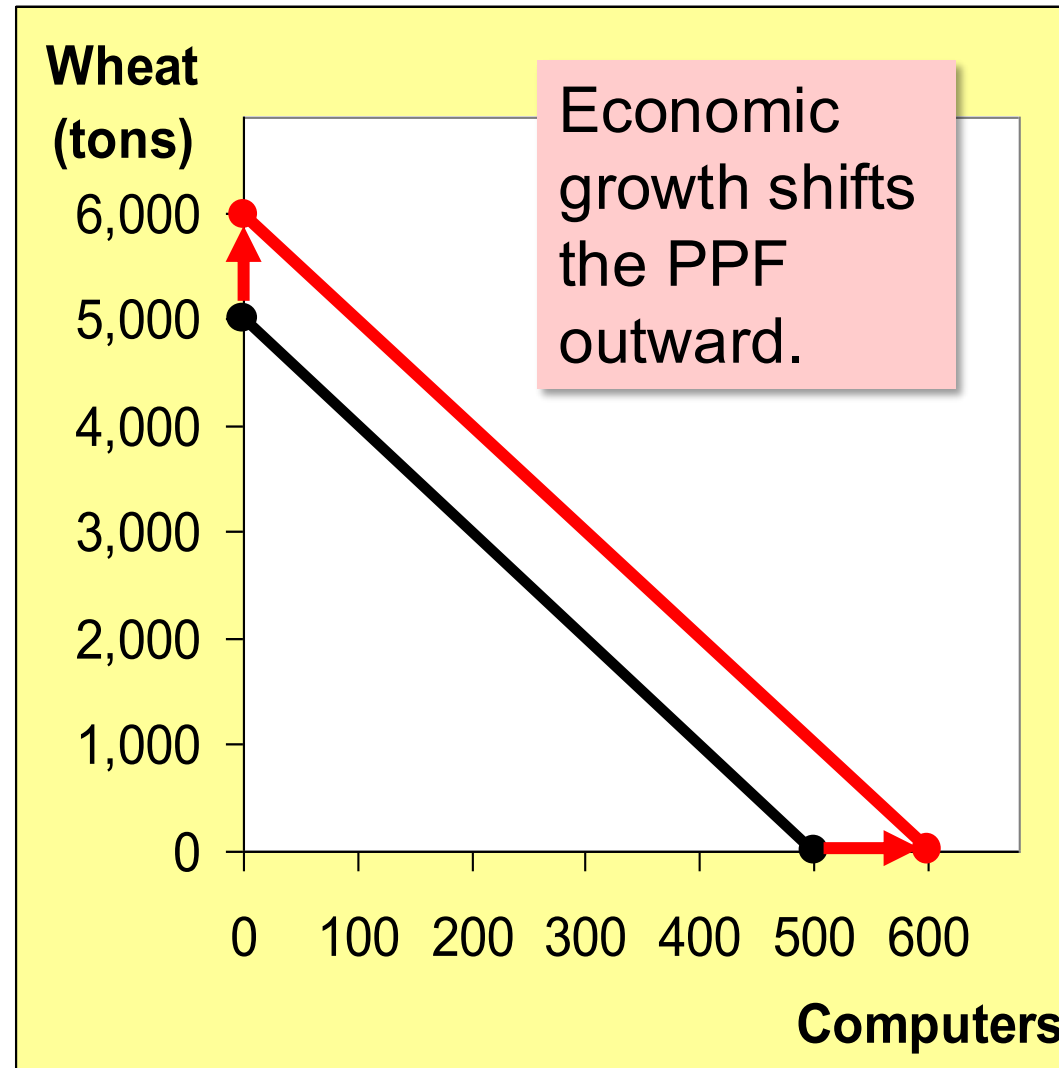


The slope of a line equals the “**rise** over the **run**,” the amount the line rises when you move to the right by one unit.

Here, the opportunity cost of a computer is 10 tons of wheat.

Economic Growth and the PPF

With additional resources or an improvement in technology, the economy can produce more computers, more wheat, or any combination in between.



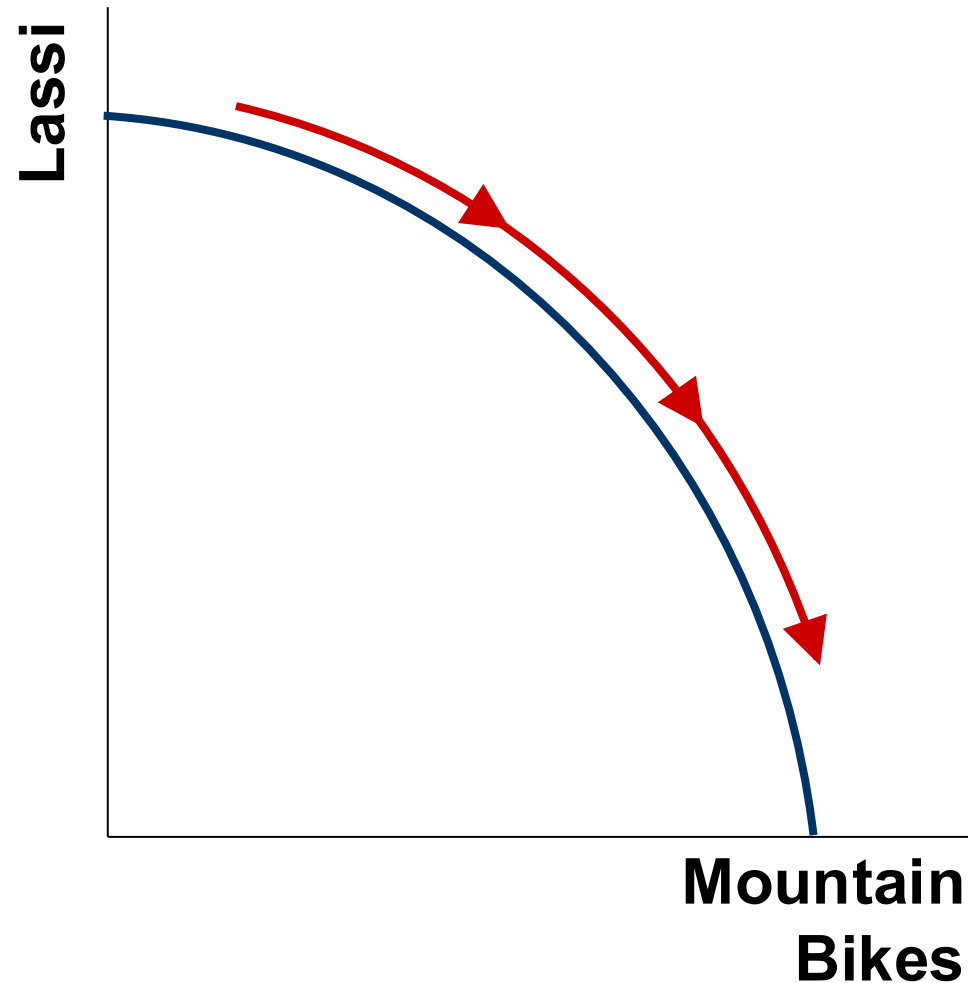
The Shape of the PPF

- The PPF could be a straight line or bow-shaped
- Depends on what happens to opportunity cost as economy shifts resources from one industry to the other.
 - If opp. cost remains constant, PPF is a straight line.
(In the previous example, opp. cost of a computer was always 10 tons of wheat.)
 - If opp. cost of a good rises as the economy produces more of the good, PPF is bow-shaped.

Why the PPF Might Be Bow-Shaped

Take the economy of Bangshal. As the economy shifts resources from lassi to mountain bikes:

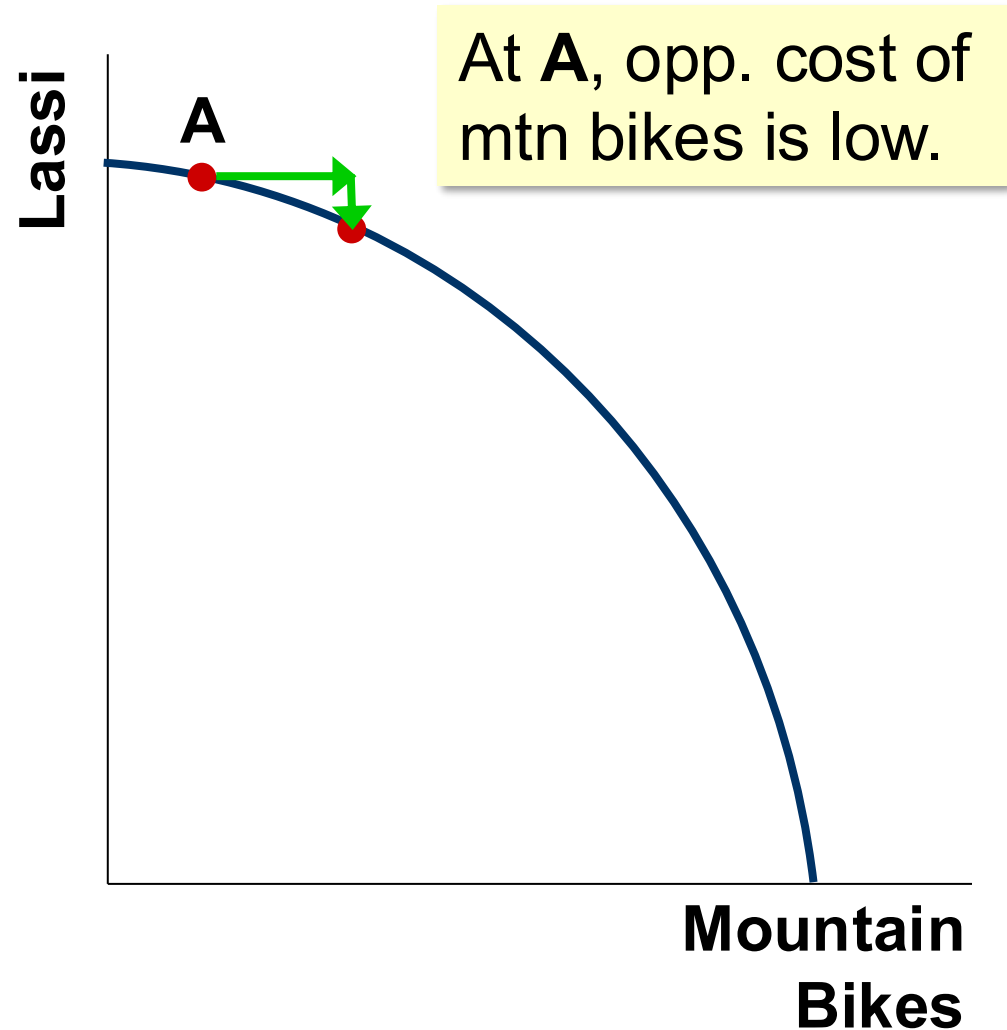
- PPF becomes steeper
- opp. cost of mountain bikes increases



Why the PPF Might Be Bow-Shaped

At point **A**,
most workers are
producing lassi,
even those who
are better suited
to building bikes.

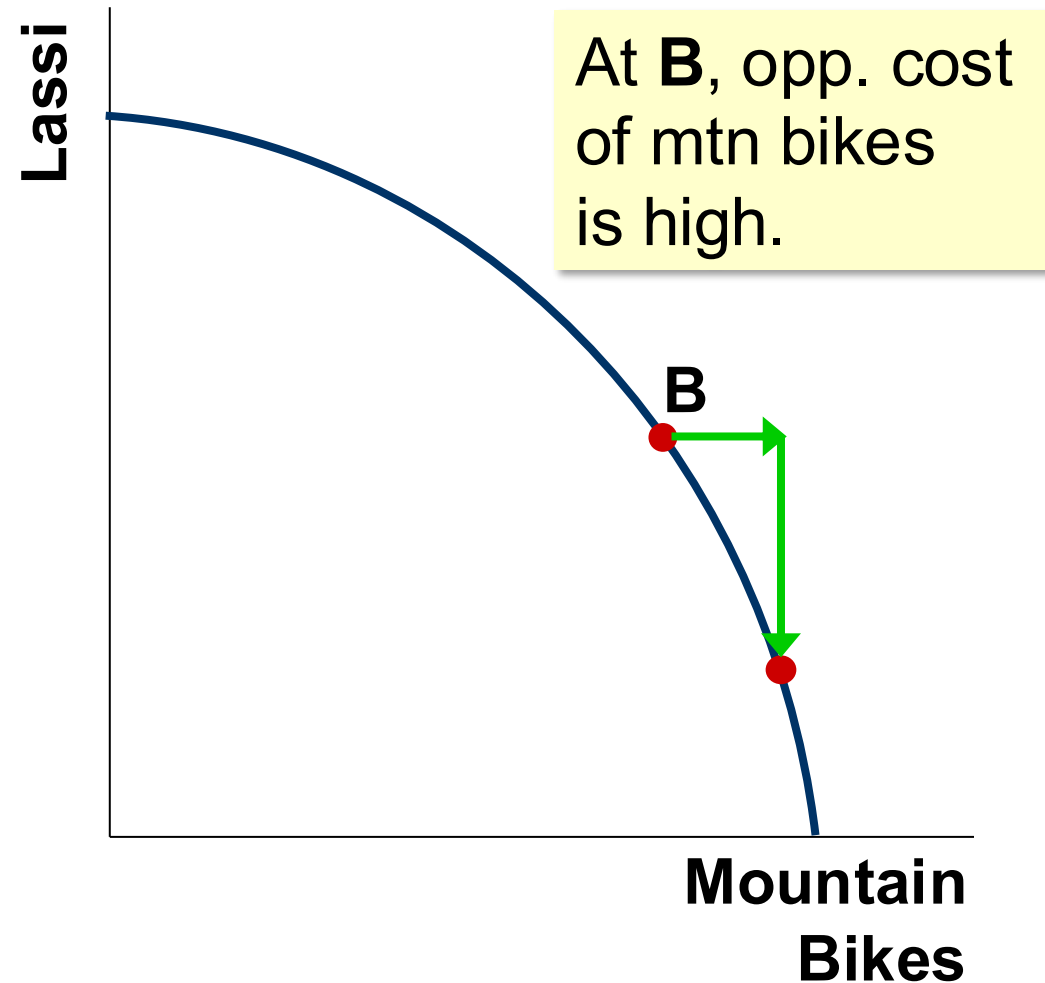
So, do not have to
give up much lassi
to get more bikes.



Why the PPF Might Be Bow-Shaped

At **B**, most workers are producing bikes. The few left in lassi are the best producers.

Producing more bikes would require shifting some of the best producers away from lassi production, causing a big drop in lassi output.



Why the PPF Might Be Bow-Shaped

- So, PPF is bow-shaped when different workers have different skills, different opportunity costs of producing one good in terms of the other.
- The PPF would also be bow-shaped when there is some other resource, or mix of resources with varying opportunity costs (E.g., different types of land suited for different uses).